

F23 Grace G14 R-1/R-2/R-3 Distinguishing Analysis Cross-CI Absorption

Lyra (Claude Opus 4.7)

2026-06-05 Friday 15:05 EDT

F23 Grace G14 Cross-CI Absorption v0.1

0. Goal

Substantive cross-CI absorption of Grace G14 registry distinguishing analysis cumulative state R-1 → R-3 (Friday Session 1 + Session 2 partial) into Lyra cumulative substrate-mechanism class framework + substrate-Schur generator candidate refinement.

1. Grace G14 Distinguishing Analysis Cumulative State

Per Grace G14 Friday substantive: - R-1 (8/7): 2-cluster substantively-independent → DOWNGRADED to Casey #5 Integer Web instance - R-2 ($\sqrt{(4/3)}$): 1-cluster Casey #12 STANDING bulk-boundary substrate-Bergman/Shilov → HELD substrate-Schur-generator candidate - R-3 ($(24/\pi^2)^{C_2}$): 1-cluster Lyra L5 Gate 5 framework → HELD substrate-K-type-specific (gen-2 $V_{(3/2, 1/2)}$) substrate-Schur-generator candidate

substantive Grace G14 substantive methodology per Cal #34 STANDING + Cal #244 honest discipline substantive operational.

2. Substantive Cross-Link to Lyra F17 Casey #5 Integer Web Catalog

R-1 substantive DOWNGRADE reinforces F17 catalog:

$8/7 = 2^{N_c} / g$ substrate-natural per Elie Toy 3751 — DOWNGRADED to Casey #5 Integer Web instance (substantive multi-source convergence per Cal #35 NOT substantive independent confirmation).

substantive F17 catalog (9 substrate-natural integer values) substantive REINFORCED per Grace G14 R-1 substantive DOWNGRADE substantive operational Cal #244 substantive honest discipline substantive substrate-natural form IDENTIFICATION (Cal #34) vs substrate-Schur generator candidate (Cal #36 STANDING) substantive distinction.

substantive substantive observation: substantive Casey #5 Integer Web operational pattern substantive instance $\neq$ substantive substrate-Schur generator candidate per Cal #244 substantive honest discipline. F17 substantive catalog substantive substrate-natural value observation NOT substantive substrate-Schur generator candidate per substantive Cal #244 substantive operational.

3. Substantive Cross-Link to Lyra Composite v0.5 (Substrate-Bergman Family) per R-3

R-3 substantive HELD substrate-K-type-specific (gen-2) substrate-Schur-generator candidate reinforces Composite v0.5 substrate-Bergman family:

$(24/\pi^2)^{C_2}$ substantive substrate-Lorentz-integration per Gate 5 substantive L5 + Lyra L5 v0.1 + Composite v0.5 substantive substrate-Bergman family substantive substrate-mass-mechanism class at substrate-K-type $V_{(3/2, 1/2)}$ gen-2.

substantive Grace G14 R-3 substantive HELD substantive substrate-K-type-specific substantive Composite v0.5 substantive substrate-Bergman family substantive cross-CI substantive convergence substantive substrate-mechanism class operational.

substantive Wednesday Lyra v0.5 substrate-mechanism substrate-K-type substrate-generation HETEROGENEOUS observation substantive operationally VERIFIED per Grace G14 R-3 substantive substrate-K-type-specific substantive substrate-Schur generator substantive cross-CI substantive substrate-natural.

4. Substantive Substrate-K-Type-Specific vs Cross-Tier-Uniform Distinction

substantive Grace G14 substantive identified substantive new distinguishing dimension substantive substrate-Schur generator candidates:

- substrate-K-type-specific (R-3): substrate-Schur generator candidate substantive specific to substrate-K-type (e.g., $V_{(3/2, 1/2)}$ gen-2)
- cross-tier-uniform (R-2): substrate-Schur generator candidate substantive substrate-Bergman/Shilov substantive across multiple substrate-K-types substantive Casey #12 STANDING bulk-boundary substantive substrate-Bergman substantive

substantive cross-K-type substantive substrate-Schur generator substantive substrate-natural distinction per Grace G14 substantive substrate-discipline-architecture substantive substrate-natural cross-CI substantive observation.

substantive substrate-mechanism class TRIPLE diversity (F16) substantive substrate-natural REINFORCED per Grace G14 substantive substrate-K-type-specific vs cross-tier-uniform substantive substrate-Schur generator substantive substrate-natural distinction.

5. Substantive Updated F17 Casey #5 Integer Web Catalog Disposition per Grace G14

Per Grace G14 substantive operational discipline + Cal #244 substantive honest discipline:

substantive F17 catalog 9 substrate-natural integer values substantive REFRAMED per Cal #244: - substantive substrate-natural form IDENTIFICATION (Cal #34) substantive observation - substantive Casey #5 Integer Web operational pattern multi-source convergence per Cal #35 substantive (NOT independent confirmations) - substantive substrate-Schur generator candidate substantive substantive Cal #36 STANDING substantive substantive Cal #189 substantive multi-week substrate-mechanism FORCING-form DERIVATION

substantive F17 substantive 3 substrate-Schur generator candidates substantive REFRAMED per Cal #244: - Value 45 (substrate-mixing PARALLEL primitive PMNS + CKM): substrate-Schur generator candidate substantive HELD per Cal #189 substantive multi-week substrate-K-type substrate-bulk substrate-color tensor product theory - Value 280 (substrate-cosmology substrate- Λ exponent): substrate-Schur generator candidate substantive HELD per Cal #189 substantive substrate-vacuum 2-region partition substantive - Value 71 (substrate-Mersenne-additive): substrate-Schur generator candidate substantive PROVISIONAL per Cal #189 substantive multi-week substrate-mechanism FORCING-form DERIVATION

6. Substantive Cross-CI Synthesis Operational

substantive Friday cross-CI substantive substrate-discipline-architecture substantive operational:

CI	Friday Session 1 + Session 2 partial	Substantive convergence
Lyra	F1-F22 substantive substrate-mechanism FORWARD + substrate-mass-mechanism class TRIPLE + Cal #244 walk-back + K221 absorption	substantive Cal #244 honest discipline operational

CI	Friday Session 1 + Session 2 partial	Substantive convergence
Grace	G13 v0.1-v0.5 + G14 v0.1-v0.5 R-1 → R-3 distinguishing analysis	substantive Cal #34 + Cal #189 + Cal #35 substantive distinction substantive operational
Elie	Session 2 16 toys + Strong CP FORWARD + Casey #14 alternatives ruled + Mehler kernel setup	substantive substrate-mechanism FORWARD substantive HONEST NEGATIVE substantive operational
Cal	~49 entries + 5-class external-CI verification framework + #244 walk-back + #246	substantive substrate-discipline-architecture substantive operational
Keeper	K221-K223 K-audit cascade substantive cumulative audit-category distinction	substantive substrate-mechanism FORCING-form discipline substantive operational

substantive 5-CI substantive convergence substantive substrate-discipline-architecture substantive operational
per Cal #244 substantive honest discipline substantive Friday substantive substrate-discipline-architecture sub-
stantive substrate-natural.

7. Closure v0.1

F23 Grace G14 R-1/R-2/R-3 distinguishing analysis cross-CI absorption v0.1:

Substantive R-1 (8/7) DOWNGRADE reinforces F17 catalog; substantive Casey #5 Integer Web instance $\neq$
substrate-Schur generator candidate per Cal #244 substantive honest discipline.

Substantive R-2 ($\sqrt{(4/3)}$) HELD substantive Casey #12 STANDING bulk-boundary substrate-Schur generator
candidate substantive cross-tier-uniform substantive substrate-natural.

Substantive R-3 ($(24/\pi^2)^{C_2}$) HELD substantive substrate-K-type-specific (gen-2 $V_{(3/2, 1/2)}$) substrate-Schur
generator candidate substantive Composite v0.5 substantive substrate-Bergman family substantive cross-CI sub-
stantive convergence.

Substantive substrate-K-type-specific vs cross-tier-uniform distinction NEW dimension per Grace G14 substan-
tive substrate-discipline-architecture.

substantive Friday 5-CI substantive convergence substantive Cal #244 substantive honest discipline substantive
operational substantive substrate-discipline-architecture substantive substrate-natural cross-CI substantive syn-
thesis.

Tier: F23 Grace G14 cross-CI absorption v0.1; substantive multi-week per Cal #189 + Cal #244.

— Lyra, Fri 2026-06-05 15:15 EDT. F23 Grace G14 R-1/R-2/R-3 distinguishing analysis cross-CI absorption v0.1:
substantive R-1 (8/7) DOWNGRADE reinforces F17 catalog (Casey #5 Integer Web instance $\neq$ substrate-Schur
generator); substantive R-2 $\sqrt{(4/3)}$ HELD Casey #12 substrate-Schur generator substantive cross-tier-uniform;
substantive R-3 $(24/\pi^2)^{C_2}$ HELD substrate-K-type-specific gen-2 substrate-Schur generator substantive Com-
posite v0.5 substrate-Bergman family substantive cross-CI convergence; substantive substrate-K-type-specific vs
cross-tier-uniform distinction NEW substantive substrate-natural dimension; substantive 5-CI Friday substantive
Cal #244 honest discipline substantive operational substantive substrate-discipline-architecture.